

Random-cluster edge correlations on graphs with at most six vertices

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Abstract

We prove that every two distinct edge indicators are nonpositively correlated in the homogeneous random-cluster model on every connected simple graph with at most six vertices, for all real $0 < p < 1$ and $0 < q < 1$. Writing $v = p/(1-p)$, we certify that each covariance numerator has the form $(q-1)B(q, v)$ with $B \in \mathbb{Z}_{\geq 0}[q, v]$. The computer-assisted proof checks all 4,208 edge pairs in 143 graph representatives. Exhaustive labelled enumeration verifies completeness, and separate exact implementations verify the polynomial identities. The accompanying certificates cover the whole parameter domain without numerical sampling. No extension to arbitrary graph order, unequal edge activities, or negative association is asserted.

1 Statement and context

Let $G = (V, E)$ be a finite connected undirected simple graph, and put $n = |V|$, $m = |E|$. For $A \subseteq E$, let $\kappa_G(A)$ count the components of the spanning subgraph (V, A) , including isolated vertices. For real $0 < p, q < 1$, define

$$\mu_{G,p,q}(A) = \frac{p^{|A|}(1-p)^{m-|A|}q^{\kappa_G(A)}}{Z_{p,q}}, \quad Z_{p,q} = \sum_{A \subseteq E} p^{|A|}(1-p)^{m-|A|}q^{\kappa_G(A)}. \quad (1)$$

The parameter p is common to all edges. Write $\mathbf{1}_e(A) = \mathbf{1}_{\{e \in A\}}$.

Theorem 1. *For every such graph with $1 \leq n \leq 6$, every pair of distinct edges $e, f \in E$, and all real $0 < p, q < 1$,*

$$\text{Cov}_{\mu_{G,p,q}}(\mathbf{1}_e, \mathbf{1}_f) \leq 0.$$

This settles the stated six-vertex homogeneous problem. It is a finite benchmark for the wider random-cluster correlation question, which is not resolved here. The certificate statement in Lemma 2 is stronger than checking selected parameter values: it is an identity of integer polynomials with a coefficientwise nonnegative quotient.

Several nearby results require careful distinction. Wagner's survey [1, Section 5.1] reports Sokal's Potts-Rayleigh computation for K_4 when $0 < q \leq 1$; its K_6 computation concerns the I-Rayleigh forest specialization. Semple and Welsh [2] discuss correlation in graphs and matroids, including series-parallel constructions. Park's wired-tree results [3, Lemma 6.3 and Theorem 6.4] concern conditioning and separated branches. Tang and Zhang [4, Theorem 1.4] study uniform connected

spanning subgraphs of sufficiently large complete graphs. These graph classes or measures differ from the complete homogeneous six-vertex layer considered here.

The four marked-edge states and division by $1 - q$ also appear in Nguyen and Pylyavskyy [5, Section 2.1]. Their subtraction-free formula is for a normalized expression evaluated at $q = 1$; their general- q representation is conjectural. We claim no innovation in this normalization. The contribution here is the complete finite collection of nonnegative coefficient certificates and its exact verification for $v > 0$ and $0 < q < 1$. As adjacent computational methodology, Zeng's finite verification of a line-graph signature inequality [6] likewise uses exhaustive graph classes and separate exact checks; that work is not a premise about random-cluster measures. The bounded primary-source comparison underlying this note did not locate an equivalent complete six-vertex statement in the inspected texts. Historical priority, including unpublished computations, is unestablished.

2 The algebraic certificate

Put $v = p/(1 - p) > 0$, and define

$$\begin{aligned} Z(q, v) &= \sum_{\substack{A \subseteq E \\ e \in A}} q^{\kappa_G(A)} v^{|A|}, & Z_e(q, v) &= \sum_{\substack{A \subseteq E \\ e \in A}} q^{\kappa_G(A)} v^{|A|}, \\ Z_f(q, v) &= \sum_{\substack{A \subseteq E \\ f \in A}} q^{\kappa_G(A)} v^{|A|}, & Z_{ef}(q, v) &= \sum_{\substack{A \subseteq E \\ e, f \in A}} q^{\kappa_G(A)} v^{|A|}. \end{aligned}$$

Here subscripts indicate restrictions on the summation, not derivatives. The common positive factor $(1 - p)^m$ in (1) cancels, so

$$\text{Cov}_{\mu_{G,p,q}}(\mathbf{1}_e, \mathbf{1}_f) = \frac{C_{G,e,f}(q, v)}{Z(q, v)^2}, \quad C_{G,e,f} = ZZ_{ef} - Z_e Z_f. \quad (2)$$

The denominator is strictly positive for $q, v > 0$.

For $a, b \in \{0, 1\}$, let

$$W_{ab} = \sum_{\substack{A \subseteq E \\ \mathbf{1}_e(A)=a, \mathbf{1}_f(A)=b}} q^{\kappa_G(A)} v^{|A|}.$$

These sums retain the activity factors of both marked edges. Expanding $Z = \sum_{a,b} W_{ab}$, $Z_e = W_{10} + W_{11}$, $Z_f = W_{01} + W_{11}$, and $Z_{ef} = W_{11}$ gives

$$C_{G,e,f} = W_{00}W_{11} - W_{10}W_{01}. \quad (3)$$

At $q = 1$ the measure factors over edges, hence $C_{G,e,f}(1, v) = 0$. Thus $q - 1$ divides $C_{G,e,f}$ in $\mathbb{Z}[q, v]$. Divisibility alone does not determine the sign of the quotient.

Lemma 2 (Finite coefficient certificate). *For every connected simple graph G of order at most six and every distinct $e, f \in E(G)$, there is a polynomial $B_{G,e,f} \in \mathbb{Z}_{\geq 0}[q, v]$ such that*

$$C_{G,e,f} = (q - 1)B_{G,e,f}. \quad (4)$$

The finite verification of this lemma is described completely in Section 3; its coefficient data and executable checker accompany the note.

Proof of Theorem 1 from Lemma 2. For $q, v > 0$, every monomial of $B_{G,e,f}$ is nonnegative. Since $q - 1 < 0$ on the required domain, (4) is nonpositive. Equation (2), with its positive denominator, proves the assertion. The map $p \mapsto p/(1 - p)$ covers all $v > 0$. Graphs with fewer than two edges have no pair to check. $\square$

Two elementary examples fix the normalization. For a triangle and any two marked edges,

$$W_{00} = q^3 + q^2v, \quad W_{10} = W_{01} = q^2v + qv^2, \quad W_{11} = qv^2 + qv^3,$$

and direct multiplication gives

$$C = (q - 1)(q^3v^3 + q^2v^4).$$

For a tree, $\kappa_G(A) = n - |A|$, and the weights factor over edges; all pair covariances are zero.

3 Exhaustive exact verification

3.1 Complete graph and edge-pair coverage

The supplied graph list contains 143 representatives. Its completeness is checked independently of the graph generator. For fixed n , list the $N = \binom{n}{2}$ possible edges lexicographically. Integers from 0 to $2^N - 1$ then encode all labelled simple graphs exactly once. A reachability search from vertex 0 identifies precisely the connected masks. Separately, apply all $n!$ vertex permutations to each supplied representative and collect the resulting masks. The checker requires that distinct representative orbits are disjoint and that their union is exactly the connected-mask set. The counts are in Table 1.

Table 1: Exact coverage. Edge pairs are counted within representatives, without an edge-pair symmetry reduction.

n	All labelled	Connected labelled	Representatives	Edge pairs
1	1	1	1	0
2	2	1	1	0
3	8	4	2	4
4	64	38	6	43
5	1,024	728	21	364
6	32,768	26,704	112	3,797
Total	33,867	27,476	143	4,208

Every graph isomorphism sends configurations and marked edges bijectively to their relabelled versions while preserving $|A|$ and $\kappa_G(A)$. It therefore preserves all partition functions used in (2). One graph per isomorphism class suffices. Within each representative, the checker constructs every pair of edge indices $i < j$ from its lexicographically sorted edge list. The set of certificate indices must equal this constructed set exactly; missing and duplicate records are rejected.

The graph6 encoding is unambiguous in this range. The first character's code minus 63 gives n . Subsequent characters, also minus 63, give six bits each, most significant first. Successive bits describe edges $\{a, b\}$ in the order $b = 1, \dots, n - 1$ and $a = 0, \dots, b - 1$. Readers check length, character range and zero padding, then sort the decoded edges lexicographically. The explicit edge list in each certificate must agree with this decoding.

3.2 Integer coefficients and a replay checker

For each representative, the producer enumerates every one of its 2^m edge subsets. A union-find component count starts with n singleton components and merges two whenever a selected edge joins distinct roots. It constructs Z, Z_e, Z_f, Z_{ef} , multiplies polynomial coefficient dictionaries, and divides C by $q - 1$ with integer synthetic division. The certificate stores C and B as sorted triples (q -degree, v -degree, coefficient), omitting zero coefficients. All stored coefficients of B must be positive.

A separate replay checker imports no producer code. It enumerates the subsets afresh and obtains components by Boolean transitive closure. Initially each vertex reaches itself and its selected-edge neighbours. Processing a possible middle vertex adds all paths through it. After all vertices are processed, the rows are exactly the reachability sets. In an undirected graph two rows coincide precisely when their vertices are in the same component; the number of distinct rows is therefore $\kappa_G(A)$. Isolated vertices retain their singleton rows.

The checker forms the four W_{ab} of (3) and uses exact integer packing to multiply them. Put

$$s = 2m + 1, \quad R = 2^{2m+1}, \quad \mathcal{E} \left(\sum_{a,b} c_{ab} q^a v^b \right) = \sum_{a,b} c_{ab} R^{as+b}.$$

In either product $W_{00}W_{11}$ or $W_{10}W_{01}$, the v -degree is at most $2m < s$, so different exponent pairs occupy different base- R positions. Each coefficient counts some of the ordered pairs of edge subsets, and is at most $2^{2m} < R$. Consequently multiplication of the encoded nonnegative integers has no carries between coefficients. The q -degree is at most $2n$, which bounds the digits to decode. Decoding the two products separately and then subtracting coefficient arrays recovers C exactly; signed borrowing never enters the encoding.

For every record the checker compares this reconstructed C with the stored coefficient list. It verifies nonnegativity of B , forms $qB - B$ coefficient by coefficient, and requires equality with C . It also checks degree bounds, exponent ordering and the absence of trailing nonzero digits. All mathematical operations use arbitrary-precision integers, with no floating-point sign decisions.

The complete replay verifies all 4,208 identities and 118,025 positive stored terms of B . There are 638 zero polynomials and 3,570 nonzero polynomials. The corresponding covariances are respectively identically zero and strictly negative throughout $0 < p, q < 1$. These counts describe the representative records; no further graph-theoretic equality classification is claimed. Complete coverage, these exact coefficient checks and isomorphism invariance prove Lemma 2.

3.3 Additional verification and its limits

A further separately written verifier recomputed every coefficient using ordinary integer convolution and descending division by $q - 1$. It computed component counts as n minus the binary incidence-matrix rank and checked them against depth-first search on all 33,867 labelled graphs. The rank identity follows componentwise: a connected component on t vertices has incidence span of dimension $t - 1$, as seen from the even-parity constraint and a spanning tree's independent columns. This implementation agreed on all identities, all coverage counts and all zero/nonzero records.

As an additional normalization check, it evaluated the original weights at $p = q = 1/2$ for every pair. Multiplication of each weight by 2^{m+n} gives the integer $2^{n-\kappa_G(A)}$. The resulting direct covariance numerator agreed with $2^{2n}C(1/2, 1)$ in every case. These rational-point checks supplement, rather than establish, the identities on the full parameter domain. Deliberately corrupted coefficients, an inadmissible negative coefficient, missing and duplicate pairs, and an omitted representative were also rejected by the replay tests.

4 Reproducibility and scope

The accompanying `supplement.zip` supplies the graph list, the full polynomial table, the producer and the independent replay checker, together with instructions and saved exact summaries. The replay requires only Python 3.9 or later and its standard library; no graph-generation package is needed. After unpacking the supplement, run the following from its root to create separate verification reports:

```
mkdir -p verification
python3 -B evidence/replay_certificates.py \
  --report verification/replay.json \
  --coverage-report verification/coverage.json
```

The supplied graph list does not serve as an assumed catalogue: the command exhausts all labelled masks to verify its completeness. The algorithmic bounds above explain why a successful exact replay certifies the entire real parameter domain, rather than an empirical sample.

Theorem 1 concerns connected simple input graphs with one common edge activity. Disconnected spanning configurations are included in every sum. Unequal activities, larger input graphs, and negative association of arbitrary increasing observables are outside the proved scope. No general random-cluster conjecture or external inequality is used in the proof. The verification is computer-assisted and has not been formalized in a proof assistant.

Computational and AI assistance. OpenAI Codex assisted with the polynomial-certificate construction, verification code, mathematical exposition and manuscript preparation. A separate model context inspected the argument and ran a separately written exact verifier before this manuscript was prepared. Separate implementations and contexts provide reproducible checks; they do not constitute external peer review or statistically independent evidence. The theorem rests on the stated finite enumeration and exact arithmetic.

References

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